

Data Science

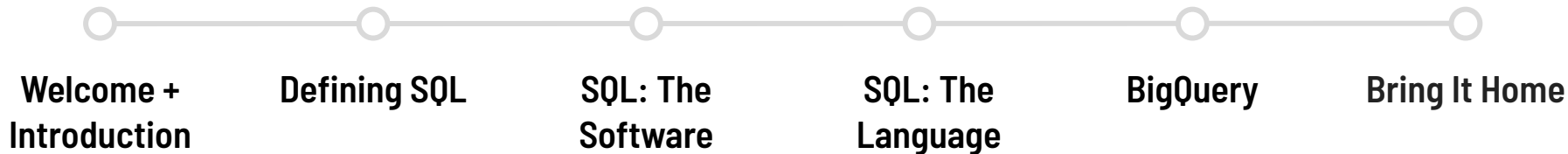
SQL I – Intro to Queries and BigQuery

Data Science Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Explain what SQL is and understand what a relational database is.

2

Use Google BigQuery to access data via SQL.

3

Write and execute a SQL query, including SELECT, FROM, LIMIT, and DISTINCT.

4

Use GROUP BY in SQL to group rows of data, and summarize those rows.

5

Practice writing aggregate functions: MIN, MAX, SUM, AVG, and COUNT.

Defining SQL

What is SQL?

Structured Query Language (SQL) is:

- Pronounced either “sequel” or “ess-kyu-ell”.
- Both the name of a language and a database software.

SQL: The Software

SQL The Database Software

A **SQL server** is a server that accepts **SQL queries** and stores data in a **SQL database**.

A SQL database is made up of many **SQL tables**.

But we already know how to store data in files!

What's the advantage of having data stored in a SQL database?

SQL ADVANTAGES

A SQL database is...

- **Relational**

- A relational database organises tables so they relate to one another via **keys**. By **joining** tables on these shared keys, you can easily query and retrieve precisely the information you need from multiple tables.

SQL ADVANTAGES

A SQL database is...

- **Relational**
 - A relational database organises tables so they relate to one another via **keys**. By **joining** tables on these shared keys, you can easily query and retrieve precisely the information you need from multiple tables.
- **Secure** (as in, “foolproof”)
 - **DBAs (database administrators)** can set up permissions structures in SQL so that the intern doesn’t accidentally delete important tables.

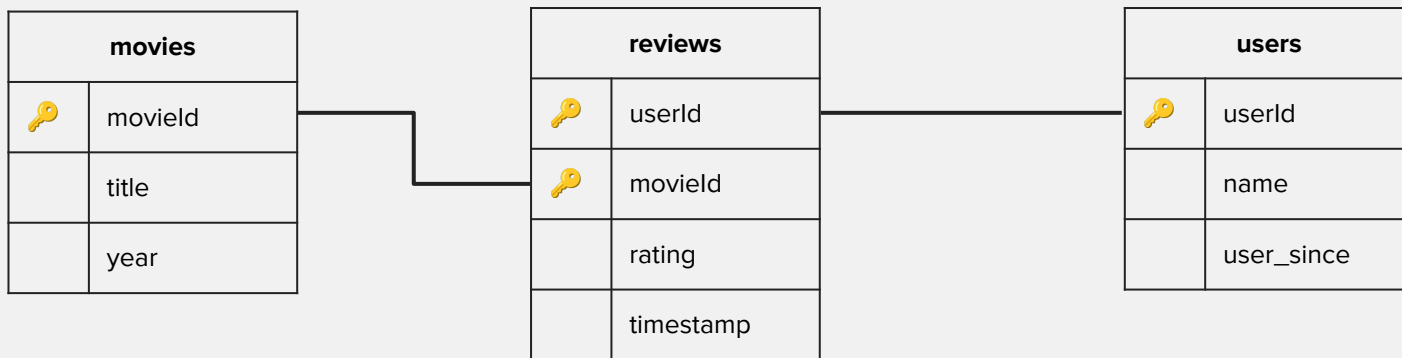
SQL ENTITY RELATIONSHIP DIAGRAMS (ERDs)

ERDs enable you to describe the relationship between tables in your database.

Again, if things are set up properly, an **entity relationship diagram (ERD)** can be made (or generated!).

This is a graph that describes the relationships between tables in your SQL DB.

ERD Example



SQL Join Example

userId	movielid	rating	timestamp
1	1	5	1/1/22 1:00
2	1	4	1/1/22 2:00
2	2	2	1/2/22 11:00

userId	name	user_since
1	Tim Book	2/28/2019
2	Katie Sylvia	8/17/2020

movielid	title	year
1	Shrek	2001
2	The Hulk	2008

userId	movielid	rating	timestamp	name	user_since	title	year
1	1	5	1/1/22 1:00	Tim Book	2/28/2019	Shrek	2001
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SQL Advantages

A SQL database is...

- ❑ **Redundant** (as in “safe”)
 - DBAs can set SQL DBs to be periodically backed-up in case something goes wrong.
 - DBs are often stored in redundant hardware on cloud servers so that if one goes down or is destroyed, data aren't lost.

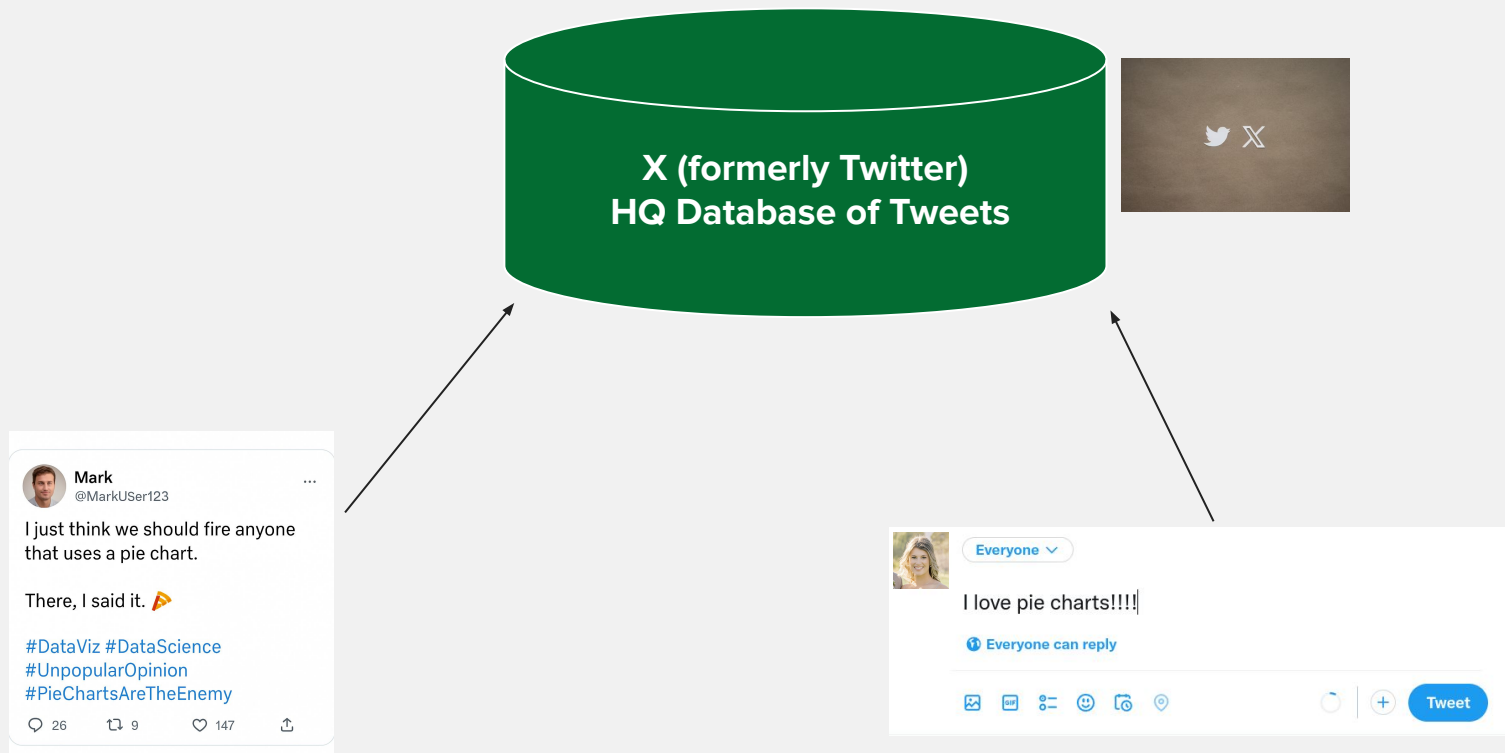
SQL Advantages

A SQL database is...

□ Parallel

- SQL DBs are designed so that multiple users can edit and query simultaneously.
- SQL can also **lock** rows in a table when one person is editing it so another person doesn't also try to edit it.
- DBs are often large in terms of storage and powerful in terms of CPU strength. Depending on the architecture, SQL can handle large, distributed calculations!

Parallel SQL Operations



SQL: The Language

SQL is also the programming language that is used to query a SQL database!

- It is a **domain-specific**, **declarative** language
 - domain-specific = only good for one domain (data stuff)
 - declarative = specify only *what* is to be done, not *how*
- As far as languages go, it is very old (1970s!)
- It is very easy to pick up. You will be able to handle 99% of queries after a few lessons

How is the SQL Language used?

- Historically it has been used as a **standalone language** to extract data from a database to then be used in reports and elsewhere
- It is also thriving as a language invoked via **glue languages** such as Python, R, JavaScript and Ruby
- SQL is the industry-standard for extracting data from relational databases

The SQL Language is not just one language...

- While there is a “SQL standard”, different versions of SQL do things pretty differently sometimes.
- These differences aren’t worth memorizing... but they’re worth Googling when you encounter something that should work but doesn’t!

There are many “dialects” of SQL

Here are just a few:

- ❑ **Postgres**: Open source, probably the most common
- ❑ **MySQL**: Also open source, but distributed by Oracle
- ❑ **Transact-SQL (T-SQL)**: SQL built by Microsoft for MS SQL Server
- ❑ **SQLite3**: A light and portable version of SQL often used to build small examples (often in Python). Contained in the Python stdlib!
- ❑ **HiveQL (HQL)**: Ok technically not a SQL dialect, but it looks pretty identical to SQL and actually powers the backend of **Spark**!

BigQuery

Google BigQuery

Google BigQuery is a **powerful cloud-based data warehouse and analytics platform**. It is designed to handle large, distributed datasets at scale, allowing users to store and analyze massive amounts of data efficiently.

SQL is used to interact with BigQuery. Specifically, BigQuery supports Standard SQL dialect (also known as GoogleSQL) for querying data.



Google
Big Query

Why Choose BigQuery?

More importantly for us today, Google provides many example databases for us to use for free! And it is a commonly used SQL solution for Data Scientists

But other advantages of BigQuery include:

- ☐ Fully managed serverless data warehouse - No infrastructure to manage
- ☐ Scalability - Handles petabyte-scale datasets effortlessly
- ☐ Speed - Query terabytes in seconds, petabytes in minutes
- ☐ Cost-effective - Pay only for storage and processing used
- ☐ Built-in AI/ML capabilities - Create and deploy ML models with SQL
- ☐ Real-time analytics - Supports streaming data and real-time querying
- ☐ Multi-cloud and multi-format - Works with various data types and cloud platforms
- ☐ Security and compliance - Strong encryption and access controls
- ☐ Easy integration - Connects seamlessly with other Google Cloud services
- ☐ Familiar SQL interface - Use standard SQL for querying and analysis

BigQuery Set Up

- Let's set up!
- You will need a Google account.
- Open **bigquery-setup-guide.md**



In SQL, the order of clauses matters

Your SQL clauses have to be in a specific order, otherwise you'll throw an exception.

1. **SELECT** - What columns do you want?
2. **FROM** - Which table?
3. **WHERE** - What conditions must be met? *haven't covered this yet*
4. **GROUP BY** - What groups do you want to see?
5. **HAVING** - What conditions must be met for groups? *haven't covered this yet*
6. **ORDER BY** - How do you want to order the results?
7. **LIMIT** - How many rows do you want to return?

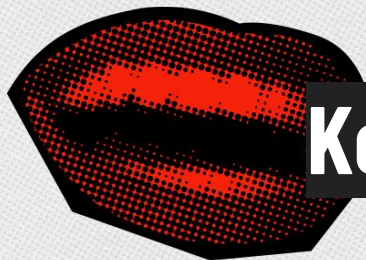
Mnemonic to remember the order: "Smelly Feet Will Give Horrible Odors, Lingeringly"

Google BigQuery

Let's walk through the repo!



Bring It Home



Key takeaways



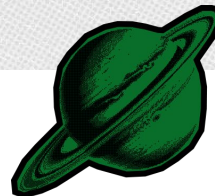
A SQL database is...

...relational, secure, redundant
and parallel



The SQL language...

...is a domain-specific,
declarative language
...is either standalone or a glue
language
...has many “dialects”



BigQuery...

...is a powerful Google
cloud-based data warehouse
and analytics platform
...uses GoogleSQL for querying
data

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DON'T FORGET:

Exit tickets



